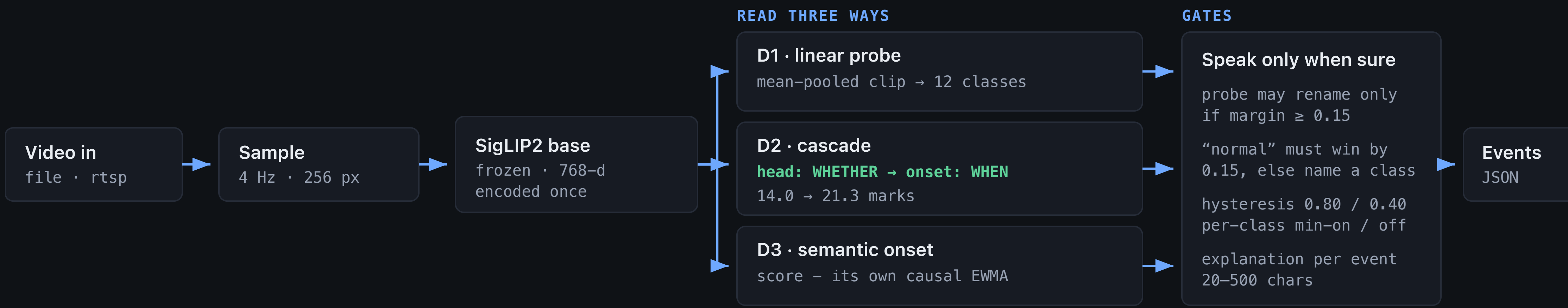
SLIDE 1 · WHAT WE BUILT

Real-time video anomaly detection — 11 classes, open-set

One frozen encoder, encoded once, read three ways. No backbone fine-tuning.

52.6	18–58×	125 s	2.4 MB	A2000
benchmark score	realtime	all 28 videos	trained weights	12 GB GPU

PIPELINE



WHY THIS SHAPE

- **Frozen encoder, one pass.** Features computed once, read three ways — why it beats realtime on a modest GPU.
- **Prompts are data, not weights.** A twelfth class is a sentence, not a retrain.
- **Score against normal.** Margin over the best of ~70 normal prompts; raw similarity just describes the scene.
- **Long video needs a baseline.** D3 keys on the rise over a causal 20 s EWMA, so a busy junction isn't permanently anomalous.
- **Locating ≠ naming.** Detector says when, probe says what — 63% vs 33% on class.

TESTED AND REJECTED

TRIED	OUTCOME
Half-max boundary refinement	D3 16.9 → 8.0
Adaptive per-video D2 thresholds	D2 14.0 → 5.3

SLIDE 2 · WHAT WE LEARNED

Every gain came from routing, not from a bigger model

D2: TWO COMPONENTS, EACH DOING HALF THE JOB

	E024 (NORMAL)	E021–23	D2
head only	silent 8.75	5.25	14.0
onset only	fires 0	12.6	12.6
cascade	silent 8.75	12.6	**21.3**

```
if head.max() >= 0.80:  # head answers WHETHER
    found = to_events(...)  # onset answers WHEN
```

The head is a good detector and a poor localiser — 0/12 events matched. The onset path is the reverse: it localises well but has no notion of “nothing here”, so it fired on the one normal video and forfeited its marks.

THE SAME IDEA, EARLIER: GATE ON CONFIDENCE

```
if top == "normal" and p1 − p2 < 0.15: name the anomaly
```

One clip read `normal 0.132` vs `smoke 0.131` — a coin flip resolved as silence, forfeiting a whole video. D1 rose **46.4% → 57.4%**.

Both changes are ~15 lines total. No new weights, no retraining.

WHAT ACTUALLY FOUND THE BUGS

- **Looking at the frames.** Rendered the frame at the midpoint of every predicted event into one labelled sheet. Ten minutes, four real defects.
- A 240 s clip called `wrong_way_driving` six times — it was queued trucks on the far carriageway.
- A dashcam **crash compilation** typed `stalled_vehicle` — the “CAR CRASHES TIME” watermark was in frame.
- A correct `fire` call silently destroyed by a later change — caught only by re-rendering.

LOCAL VALIDATION NEVER PREDICTED THE SCORE

LOCAL SIGNAL	SAID	REALITY
Marks formula	29.9	51.9
Train accuracy	+27	−24
Val IoU 0.66 → 0.86	better	worse
Cross-val 33% → 58%	better	±0